

Esercizio 1. Let $\mathcal{T}$ be a p.u. theory. Then the following are equivalent:

1. $\mathcal{T}$ is irreducible
2. $\mathcal{T}$ has JEP

Proof. • 1) $\Rightarrow$ 2): Let $A, B \models \mathcal{T}$. By contradiction assume that no model of $\mathcal{T}$ admits morphisms from A, B . Equivalently, $\mathcal{T} \cup \text{Diag}^+(A) \cup \text{Diag}^+(B)$ is inconsistent.¹ By compactness we can find $\tau, \phi(\vec{a}), \psi(\vec{b})$ inconsistent, where τ is a finite conjunction of formulas in $\mathcal{T}$, $\phi(\vec{a})$ is a finite conjunction of formulas in $\text{Diag}^+(A)$ and $\psi(\vec{b})$ is a finite conjunction of formulas in $\text{Diag}^+(B)$. Equivalently $\mathcal{T} \models \neg(\phi(\vec{a}) \wedge \psi(\vec{b}))$. Since $\vec{a}, \vec{b}$ are fresh constants, this is the same as $\mathcal{T} \models \neg(\exists y \phi(y) \wedge \exists z \psi(z))$. Equivalently (because morphisms preserve pp) $\mathcal{T} \models \forall y \neg \phi(y) \vee \forall z \neg \psi(z)$. Note that this is a disjunction of p.u. so by irreducibility we can assume for example $\mathcal{T} \models \forall y \neg \phi(y)$. In particular $A \models \forall y \neg \phi(y)$. This is a contradiction since $\phi(\vec{a}) \in \text{Diag}^+(A)$.

• 2) $\Rightarrow$ 1): By contradiction, let α, β p.u. sentences such that $\mathcal{T} \models \alpha \vee \beta$ but $\mathcal{T} \not\models \alpha, \beta$. Let then A, B models of $\mathcal{T}$ such that $A \not\models \alpha, B \not\models \beta$. Let C model of $\mathcal{T}$ found using JEP on A, B . Since $C \models \mathcal{T}$ it must be $C \models \alpha$ or $C \models \beta$. Assume for example that $C \models \alpha$. Note that $\neg \alpha$ is pp and $A \models \neg \alpha$, so $C \models \neg \alpha$ (because morphisms preserve pp). Contradiction. $\square$

¹ $\text{Diag}^+(A) = \{\text{atomic formulas with parameters in } A \text{ satisfied by } A\}$ and same for B